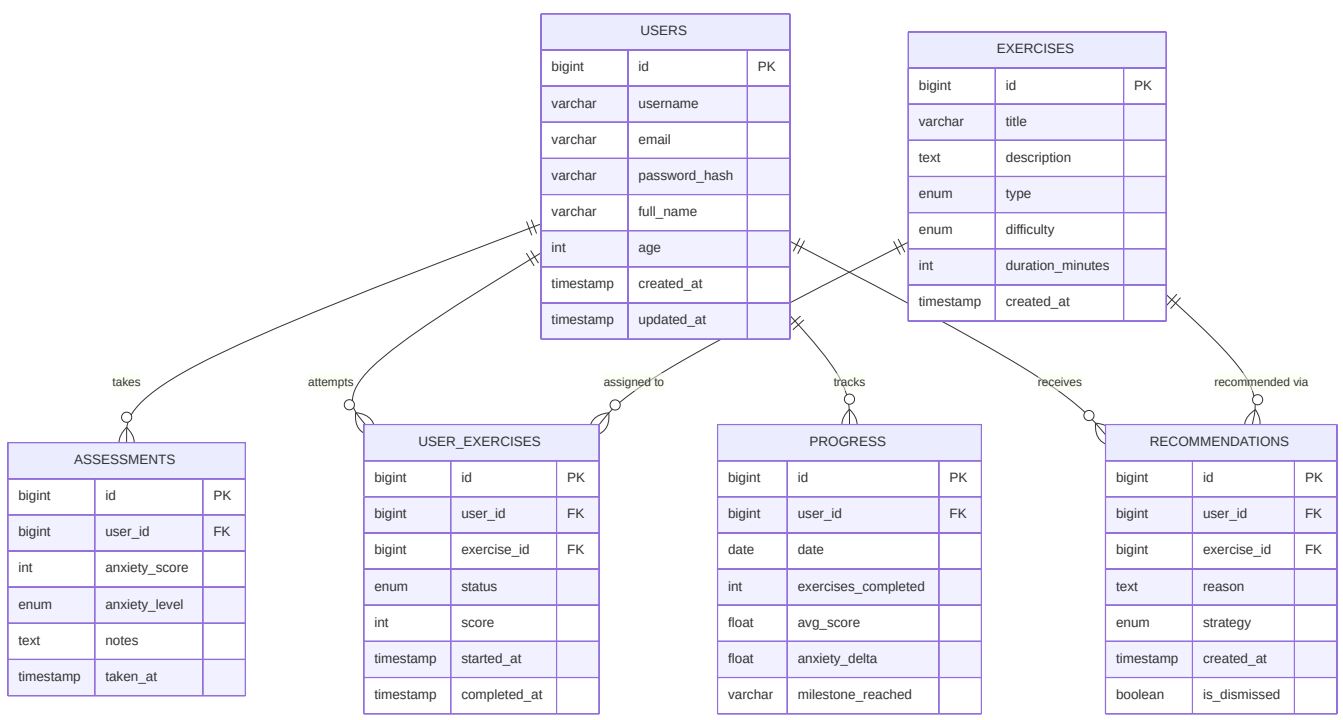


SocialEase – API Schema & Endpoints Reference

Social Anxiety Helper Web Application
Backend: Java (Spring Boot) | Database: PostgreSQL

Database Schema

Entity Relationship Diagram



users

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY
username	VARCHAR(50)	NOT NULL, UNIQUE
email	VARCHAR(100)	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	NOT NULL
full_name	VARCHAR(100)	
age	INTEGER	
created_at	TIMESTAMPTZ	DEFAULT NOW()

Column	Type	Constraints
updated_at	TIMESTAMPTZ	DEFAULT NOW()

assessments

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY
user_id	BIGINT	FOREIGN KEY → users.id
anxiety_score	INTEGER	NOT NULL CHECK (anxiety_score BETWEEN 0 AND 100)
anxiety_level	VARCHAR(10)	CHECK (anxiety_level IN ('LOW', 'MODERATE', 'HIGH'))
notes	TEXT	
taken_at	TIMESTAMPTZ	DEFAULT NOW()

exercises

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY
title	VARCHAR(150)	NOT NULL
description	TEXT	NOT NULL
type	VARCHAR(20)	CHECK (type IN ('BREATHING', 'SOCIAL_SIMULATION', 'CONFIDENCE', 'COMMUNICATION'))
difficulty	VARCHAR(15)	CHECK (difficulty IN ('BEGINNER', 'INTERMEDIATE', 'ADVANCED'))
duration_minutes	INTEGER	
created_at	TIMESTAMPTZ	DEFAULT NOW()

user_exercises

(Tracks which exercises a user has attempted or completed)

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY

Column	Type	Constraints
user_id	BIGINT	FOREIGN KEY → users.id
exercise_id	BIGINT	FOREIGN KEY → exercises.id
status	VARCHAR(15)	CHECK (status IN ('IN_PROGRESS', 'COMPLETED', 'SKIPPED'))
score	INTEGER	
started_at	TIMESTAMPTZ	
completed_at	TIMESTAMPTZ	

progress

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY
user_id	BIGINT	FOREIGN KEY → users.id
date	DATE	NOT NULL
exercises_completed	INTEGER	DEFAULT 0
avg_score	NUMERIC(5,2)	
anxiety_delta	NUMERIC(5,2)	(change from previous assessment)
milestone_reached	VARCHAR(100)	

recommendations

Column	Type	Constraints
id	BIGSERIAL	PRIMARY KEY
user_id	BIGINT	FOREIGN KEY → users.id
exercise_id	BIGINT	FOREIGN KEY → exercises.id
reason	TEXT	
strategy	VARCHAR(25)	CHECK (strategy IN ('GRADUAL_EXPOSURE', 'COGNITIVE_REFRAMING', 'MINDFULNESS'))
created_at	TIMESTAMPTZ	DEFAULT NOW()
is_dismissed	BOOLEAN	DEFAULT FALSE

REST API Endpoints

Base URL: `/api/v1`

1. User Module

Register

POST `/api/v1/auth/register`

Request Body:

```
{
  "username": "string",
  "email": "string",
  "password": "string",
  "fullName": "string",
  "age": 0
}
```

Response: `201 Created`

```
{
  "id": 1,
  "username": "string",
  "email": "string",
  "token": "jwt_token"
}
```

Login

POST `/api/v1/auth/login`

Request Body:

```
{
  "email": "string",
  "password": "string"
}
```

Response: `200 OK`

```
{
  "token": "jwt_token",
  "userId": 1,
  "username": "string"
}
```

Get User Profile

```
GET /api/v1/users/{userId}
```

Headers: Authorization: Bearer <token>

Response: 200 OK

```
{
  "id": 1,
  "username": "string",
  "email": "string",
  "fullName": "string",
  "age": 0,
  "createdAt": "timestamp"
}
```

Update User Profile

```
PUT /api/v1/users/{userId}
```

Headers: Authorization: Bearer <token>

Request Body:

```
{
  "fullName": "string",
  "age": 0
}
```

Response: 200 OK

Delete User

```
DELETE /api/v1/users/{userId}
```

Headers: Authorization: Bearer <token>

Response: 204 No Content

2. Assessment Module

Submit an Assessment

```
POST /api/v1/assessments
```

Headers: Authorization: Bearer <token>

Request Body:

```
{
  "userId": 1,
  "anxietyScore": 60,
  "notes": "Felt anxious before a meeting"
}
```

Response: 201 Created

```
{
  "id": 1,
  "userId": 1,
  "anxietyScore": 60,
  "anxietyLevel": "MODERATE",
  "takenAt": "timestamp"
}
```

Get All Assessments for a User

```
GET /api/v1/assessments/user/{userId}
```

Headers: Authorization: Bearer <token>

Response: 200 OK — Array of assessment objects

Get Latest Assessment

```
GET /api/v1/assessments/user/{userId}/latest
```

Headers: Authorization: Bearer <token>

Response: 200 OK — Single assessment object

Get Assessment by ID

```
GET /api/v1/assessments/{assessmentId}
```

Headers: Authorization: Bearer <token>

Response: 200 OK

3. Exercise Module

Get All Exercises

```
GET /api/v1/exercises
```

Query Params: `?type=BREATHING&difficulty=BEGINNER`

Response: `200 OK` — Array of exercise objects

Get Exercise by ID

```
GET /api/v1/exercises/{exerciseId}
```

Response: `200 OK`

```
{
  "id": 1,
  "title": "string",
  "description": "string",
  "type": "BREATHING",
  "difficulty": "BEGINNER",
  "durationMinutes": 5
}
```

Start an Exercise (User)

```
POST /api/v1/exercises/{exerciseId}/start
```

Headers: `Authorization: Bearer <token>`

Request Body:

```
{
  "userId": 1
}
```

Response: `201 Created`

```
{
  "userExerciseId": 1,
  "status": "IN_PROGRESS",
  "startedAt": "timestamp"
}
```

Complete an Exercise (User)

```
PUT /api/v1/exercises/{exerciseId}/complete
```

Headers: Authorization: Bearer <token>

Request Body:

```
{
  "userId": 1,
  "score": 85
}
```

Response: 200 OK

```
{
  "userExerciseId": 1,
  "status": "COMPLETED",
  "score": 85,
  "completedAt": "timestamp"
}
```

Get User's Exercise History

```
GET /api/v1/exercises/user/{userId}/history
```

Headers: Authorization: Bearer <token>

Query Params: ?status=COMPLETED

Response: 200 OK — Array of user_exercise records

4. Progress Tracker

Get Progress Summary

```
GET /api/v1/progress/user/{userId}
```

Headers: Authorization: Bearer <token>

Response: 200 OK

```
{
  "userId": 1,
  "totalExercisesCompleted": 20,
  "currentAnxietyLevel": "LOW",
  "improvementPercentage": 35.5,
  "milestones": ["First Exercise", "7-Day Streak"]
}
```

Get Daily Progress

```
GET /api/v1/progress/user/{userId}/daily
```

Headers: Authorization: Bearer <token>

Query Params: ?from=2024-01-01&to=2024-03-01

Response: 200 OK — Array of daily progress records

Get Progress by Date

```
GET /api/v1/progress/user/{userId}/date/{date}
```

Headers: Authorization: Bearer <token>

Path Param: date format YYYY-MM-DD

Response: 200 OK

```
{
  "date": "2024-02-15",
  "exercisesCompleted": 3,
  "avgScore": 78.5,
  "anxietyDelta": -5.0,
  "milestoneReached": "Weekly Goal"
}
```

5. Recommendation System

Get Recommendations for a User

```
GET /api/v1/recommendations/user/{userId}
```

Headers: Authorization: Bearer <token>

Response: 200 OK

```
[
  {
    "id": 1,
    "exercise": { "id": 3, "title": "Mirror Talk Exercise" },
    "reason": "Based on your HIGH anxiety score, this exercise helps build self-confidence.",
    "strategy": "COGNITIVE_REFRAMING"
  }
]
```

Generate New Recommendations

POST /api/v1/recommendations/generate

Headers: Authorization: Bearer <token>

Request Body:

```
{
  "userId": 1
}
```

Response: 201 Created — Array of newly generated recommendation objects

Dismiss a Recommendation

PUT /api/v1/recommendations/{recommendationId}/dismiss

Headers: Authorization: Bearer <token>

Response: 200 OK

```
{
  "id": 1,
  "isDismissed": true
}
```

HTTP Status Codes Used

Code	Meaning
200	OK – Request succeeded
201	Created – Resource successfully created
204	No Content – Successful deletion
400	Bad Request – Invalid input
401	Unauthorized – Missing or invalid token
403	Forbidden – Insufficient permissions
404	Not Found – Resource does not exist
409	Conflict – Duplicate resource (e.g., email already registered)
500	Internal Server Error